IKARUS – STAGE 3

INNOVATION CHALLENGE

Problem-Based Module (PBM) Statements

TEAM INNOVATION AND WEBSITE DEVELOPMENT

This document presents the nine PBM problem statements, their guiding questions, and the common explanation requirements applicable to all teams.

1. Smart Digital Queue Management System for Reducing Waiting Time in Public Service Centers

Problem Statement: Design and develop a website that manages digital tokens, estimates waiting times, and helps public service centers organize queues efficiently, including support for urgent cases.

Guiding Questions

1. How can users receive digital tokens and know their expected waiting time?
2. How can the website handle missed tokens, delays, and urgent cases fairly?
3. What information should staff update to keep queue status accurate?
4. How can the website reduce crowding at public service centers?

2. Electric Vehicle Charging Station Management Website

Problem Statement: Design and develop a website that helps electric vehicle users find suitable charging stations by displaying availability, charging options, and estimated waiting times.

Guiding Questions

5. How can users find available charging stations suitable for their vehicles?
6. How can the website keep station availability information accurate?
7. What details would help users compare nearby charging stations?
8. How can a prototype demonstrate the website using sample data?

3. Waterlogging Prediction and Early Warning Website

Problem Statement: Design and develop a website that identifies locations at risk of waterlogging using rainfall information, drainage conditions, and historical incidents, and displays timely warnings.

Guiding Questions

9. What data can help identify areas at risk of waterlogging?
10. How can the website display risk levels and warnings clearly?
11. How can people identify safer alternative routes during heavy rainfall?
12. How should the website handle missing or outdated information?

4. Public Transport Delay Prediction and Passenger Information Website

Problem Statement: Design and develop a website that helps passengers understand possible public transport delays and plan journeys using route information, traffic conditions, and historical travel times.

Guiding Questions

13. What information is needed to estimate possible transport delays?
14. How can passengers compare alternative routes and estimated travel times?
15. How can the website provide useful updates to people with limited internet access?
16. How should it communicate when exact arrival times are unknown?

5. Smart Medicine Stock and Expiry Management Website

Problem Statement: Design and develop a website for pharmacies and healthcare centres to monitor medicine stock, track expiry dates, and identify possible shortages or excess stock.

Guiding Questions

17. How can staff record incoming, sold, and remaining medicine quantities?
18. How can the website alert staff about medicines nearing expiry or running low?
19. How can it help staff identify urgent medicine shortages?
20. What reports would help reduce medicine wastage?

6. Smart Energy Distribution Website for Critical Facilities

Problem Statement: Design and develop a website that helps facility managers plan limited backup electricity during power outages by prioritizing essential services and monitoring available backup capacity.

Guiding Questions

21. How should essential services be prioritized when backup power is limited?
22. What information is needed about power demand and backup capacity?
23. How can the plan change if the power outage lasts longer than expected?
24. How can the website explain and record the reasons for each priority decision?

7. Smart Parking Availability and Reservation Website

Problem Statement: Design and develop a website that helps drivers find and reserve available parking spaces based on location, parking duration, and vehicle requirements.

Guiding Questions

25. How can drivers search for parking spaces that match their requirements?
26. How can the website prevent two users from reserving the same space?
27. What details should be displayed, such as parking charges, location, and time limits?
28. How can the website reduce the time drivers spend searching for parking?

8. Smart Personal Data Privacy Protection Website

Problem Statement: Design and develop a website that helps users understand the personal information requested by digital services and provides practical guidance for improving privacy and account security.

Guiding Questions

29. How can the website explain privacy permissions in simple language?
30. How can users identify requests for unnecessary personal information?
31. What steps can the website recommend to improve account privacy and security?
32. How can the website protect users' sensitive information?

9. Smart Price Difference Finder Website

Problem Statement: Design and develop a website that compares the prices of the same product across different online stores, displays price differences, and helps users identify suitable deals.

Guiding Questions

33. How can the website identify the same product across different stores when product names or descriptions differ?
34. How can it display product prices, delivery charges, availability, and price-checking dates?
35. How can it avoid misleading comparisons when product sizes or variants differ?
36. How can a working prototype demonstrate price comparison using sample data?

10. Accessible Learning Material Converter Website

Problem Statement: Design and develop a website that makes study materials easier to access by offering features such as simplified text, readable layouts, and text-to-speech support.

Guiding Questions

37. How can students upload study materials and choose an accessible format?
38. How can the website simplify text while preserving the original meaning?
39. What features can support students with different reading and learning needs?
40. How can the website handle scanned documents or materials that cannot be read clearly?

Common Instructions for All Teams

For the selected PBM statement, each team should clearly explain the following points:

- **Problem identification:** What is the real-world problem?
- **Proposed solution:** How will the website address it?
- **Key features:** What functions will the website provide?
- **Technology:** What technologies and tools will be used?
- **Expected benefits:** How will the solution help its intended users?

Team preparation checklist

Item	Completed
Selected PBM statement identified	☐
Real-world problem explained	☐
Proposed website solution described	☐
Key features listed	☐
Technology and tools identified	☐
Expected benefits explained	☐